

DATA 505 (Statistics Using R): Exam 2

Part 2: Open Computer

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Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are “not alive” (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed .qmd file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze the `MA_schools` dataset from the `moderndive` package, which contains data on Massachusetts public high schools in 2017.

```
# Install moderndive package if not already installed
if (!require(moderndive)) {
  install.packages("moderndive")
}

# Load required packages
library(moderndive)
library(ggplot2)

# Load the MA_schools dataset
data(MA_schools)
MA_schools
```

```
# A tibble: 332 x 4
  school_name      average_sat_math perc_disadvan size
  <chr>          <dbl>          <dbl> <fct>
1 Abington High      516          21.5 medium
2 Agawam High        514          22.7 large
3 Amesbury High      534          14.6 large
4 Andover High       581           6.3 large
5 Arlington High     592          10.3 large
6 Ashland High       576          10.3 large
7 Attleboro High    504          25.6 large
8 Auburn Senior High 505          15.2 large
9 Avon Middle High School 481          23.8 small
10 Barnstable High   513          25.5 large
# i 322 more rows
```

Data Exploration (3 points)

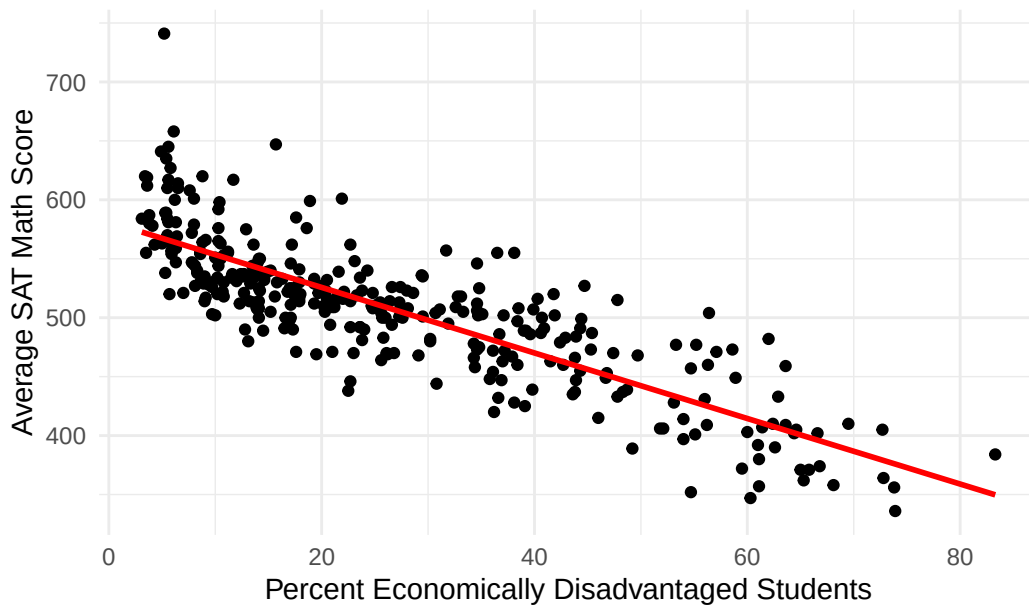
1 (3 points)

Create a scatterplot showing the relationship between `perc_disadvan` (percent economically disadvantaged students) and `average_sat_math` (average SAT math score). Based on the plot, do you expect a positive or negative relationship?

```
ggplot(MA_schools, aes(x = perc_disadvan, y = average_sat_math)) +
  geom_point() +
  labs(
    x = "Percent Economically Disadvantaged Students",
    y = "Average SAT Math Score",
    title = "Relationship Between Economic Disadvantage and SAT Math Scores"
  ) +
  geom_smooth(method = "lm", se = FALSE, color = "red", linewidth = 1) +
  theme_minimal()
```

``geom_smooth()`` using formula = 'y ~ x'

Relationship Between Economic Disadvantage and SAT Math



Answer: The scatterplot shows a negative relationship. Schools with a higher percentage of economically disadvantaged students tend to have lower average SAT math scores. The points generally slope downward from left to right.

Simple Linear Regression (12 points)

1 (4 points)

Fit a simple linear regression model predicting `average_sat_math` from `perc_disadvan`. Store the model as `model1` and display the summary.

```
model1 <- lm(average_sat_math ~ perc_disadvan, data = MA_schools)
summary(model1)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan, data = MA_schools)
```

Residuals:

Min	1Q	Median	3Q	Max
-80.74	-21.26	-4.12	18.54	174.17

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	581.2811	3.2668	177.9	<2e-16 ***
perc_disadvan	-2.7798	0.1011	-27.5	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.54 on 330 degrees of freedom

Multiple R-squared: 0.6962, Adjusted R-squared: 0.6953

F-statistic: 756.2 on 1 and 330 DF, p-value: < 2.2e-16

2 (4 points)

Extract the estimated model coefficients (intercept and slope) from this fitted model object, and store them in a vector called `sat_effects`. Then report the intercept and slope below.

```
sat_effects <- coef(model1)
sat_effects
```

```
(Intercept) perc_disadvan
581.281054    -2.779787
```

Intercept: 581.281.

Slope: -2.78.

3 (4 points)

Interpret the slope coefficient in context. What does it tell you about the relationship between the percentage of economically disadvantaged students and average SAT math scores?

Interpretation: The slope tells us how SAT math scores change when economic disadvantage increases. For each 1 percentage point increase in economically disadvantaged students, the average SAT math score decreases by the slope value, this means that as more students in a school face economic disadvantage, SAT performance tends to drop, suggesting socioeconomic conditions may negatively impact academic performance.

Making Predictions (8 points)

1 (4 points)

Use `model1` to predict the average SAT math score for a school where 50% of students are economically disadvantaged. Create a data frame with this value and use the `predict()` function.

```
new_school <- data.frame(perc_disadvan = 50)
predicted_score <- predict(model1, newdata = new_school)
predicted_score
```

```
      1
442.2917
```

Predicted average SAT math score: 442.2917.

2 (4 points)

Calculate and print out a 95% prediction interval for a single new school where 50% of students are economically disadvantaged.

```
predict(model1, newdata = new_school, interval = "prediction", level = 0.95)
```

```
      fit      lwr      upr
1 442.2917 376.0542 508.5292
```

Multiple Linear Regression (12 points)

1 (4 points)

Fit a multiple regression model predicting `average_sat_math` from both `perc_disadvan` and `size` (school size category). Store this as `model2` and display the summary.

```
model2 <- lm(average_sat_math ~ perc_disadvan + size, data = MA_schools)
summary(model2)
```

```
Call:
lm(formula = average_sat_math ~ perc_disadvan + size, data = MA_schools)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-81.342	-20.802	-3.872	18.211	167.251

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	588.1898	7.6067	77.325	<2e-16 ***
perc_disadvan	-2.7772	0.1063	-26.120	<2e-16 ***
sizemedium	-11.9134	7.5346	-1.581	0.115
sizelarge	-6.3615	6.9231	-0.919	0.359

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.5 on 328 degrees of freedom

Multiple R-squared: 0.6988, Adjusted R-squared: 0.696

F-statistic: 253.6 on 3 and 328 DF, p-value: < 2.2e-16

2 (4 points)

Compare the R-squared values of `model1` and `model2`. Which model explains more variance in average SAT math scores?

```
summary(model1)$r.squared
```

```
[1] 0.6961948
```

```
summary(model2)$r.squared
```

```
[1] 0.6987587
```

Answer: If `model2` has a higher R-squared than `model1`, then including size helps explain more variation in SAT math scores. This means school size provides additional useful information beyond economic disadvantage alone.

3 (4 points)

Use the `anova()` function to formally compare `model1` and `model2`. Does adding `size` significantly improve the model at the $\alpha = 0.05$ level?

```
anova(model1, model2)
```

Analysis of Variance Table

Model 1: `average_sat_math ~ perc_disadvan`

Model 2: `average_sat_math ~ perc_disadvan + size`

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	330	371191				
2	328	368058	2	3132.6	1.3958	0.2491

Conclusion: If the p-value < 0.05 , then adding `size` significantly improves the model, meaning school size has a statistically meaningful relationship with SAT math scores, after controlling for economic disadvantage. If the p-value > 0.05 , then adding `size` does not significantly improve model fit, meaning school size does not explain enough additional variation to be considered important at the 5% significance level..